Cascaded $\chi^{(2)}$:$\chi^{(2)} \to \chi^{(3)}_{\text{eff}}$ in a Doubly-Resonant Lithium Niobate Microring

Feasibility, observables, and comparison with a native-$\chi^{(3)}$ ring

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Abstract

We adapt the cascaded $\chi^{(2)}$:$\chi^{(2)}$ Kerr analysis of the metasurface note to a travelling-wave thin-film lithium niobate (LN) microring with loaded $Q_1 \sim 10^5$–10^6 at the fundamental. Three things change qualitatively relative to the metasurface, and all three work in the microring's favour. First, the absolute scale of the cascade is no longer a modelling unknown: the $\chi^{(2)}$ photon–photon coupling rate g is fixed by the *measured* normalised second-harmonic (SH) generation efficiency through a relation we derive and check against the published expression, so the $O(1)$ "bridge" constant that limited the metasurface sample analysis disappears. Second, because $\omega_2 = 2\omega_1$ while $Q_2 \lesssim Q_1$, the SH mode is intrinsically *broader* than the fundamental, so the cascade costs no device bandwidth — the opposite of the metasurface, where the SH linewidth was the bottleneck. Third, loaded $Q_2 \sim 3 \times 10^5$ at 780 nm is experimentally plausible with precedent, rather than exceptional. Taking the demonstrated PPLN microring coupling $g/2\pi = 1.2$ MHz and scaling it to a smaller target ring ($g/2\pi \simeq 1.4$ MHz, an extrapolation), the cascaded Kerr coefficient exceeds the native LN Kerr by $R \simeq 1.8 \times 10^3$ at the conservative operating point $\Delta\omega = 2\gamma_2$, equivalent to $n_2^{\text{eff}} \simeq -3.3 \times 10^{-16}$ m^2/W. Bistability moves from 14 mW to 32–37 μW; a multimode Bogoliubov stability calculation puts the parametric threshold at 8.4 μW against 9.4 mW natively. Without periodic poling the advantage collapses to $R \simeq 13$, so poling is the enabling step. Two design rules follow: bistability and single-pair parametric gain both require the *pump-frame* mismatch to satisfy $|\Delta_{\text{eff}}| > \frac{\sqrt{3}}{2}\gamma_2$, where $\Delta_{\text{eff}} = \Delta\omega - 2\delta_1$; and the Kerr description requires the envelope bandwidth to stay below $\sqrt{\kappa_2^2 + \delta_2^2}$, which for this device is $27\kappa_1$. The binding obstacles are thermal and photorefractive shifts, which are within a factor of two of the cascaded Kerr at the relevant powers. The cascade is *odd* in $\Delta\omega$ where they are, to leading order, *even*, and lithium niobate's own Pockels effect tunes $\Delta\omega$ by $2.4\gamma_2$ per V/μm, so an electro-optically dithered, antisymmetric-in-$\Delta\omega$ lock-in measurement suppresses them. That measurement is the recommended first experiment. The cascade is *not* a route to single-photon anharmonicity: the dispersive figure of merit is $\sim (g/\kappa_1)(g/\kappa_2)$, quadratically worse than the resonant g/κ_1.

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1 Setting and goal

The main note derives the cascaded response $\chi^{(3)}_{\mathrm{casc}}(\omega_1)$ of a doubly-resonant dielectric metasurface: a quasi-BIC at ω_1 and a high-Q mode near $2\omega_1$, with the SH adiabatically eliminated to leave a Kerr-like term on the fundamental. The same two-mode structure describes a doubly-resonant $\chi^{(2)}$ microring, and the derivation carries over essentially unchanged. What changes is the *numbers*, and they change by three orders of magnitude.

The reason is structural. The metasurface enhancement was $R \propto Q_2$ with $Q_2 \sim 4 \times 10^3$ realistically available in a quasi-BIC; the four fabricated SH-only samples and Design 6 delivered $R^{(1\mathrm{R})} = 0.4$–2.5 once the true modal overlaps were measured. A thin-film LN microring routinely reaches $Q_2 \sim 3$–6×10^5 at the SH wavelength, two orders of magnitude higher, and the modal overlap problem is solved by a different mechanism — quasi-phase matching via periodic poling,

which recovers the full d_{33} over the entire circumference.

Three questions organise what follows.

(i) **Is the platform good enough?** Section 5 answers this with the enhancement ratio R, anchored on a measured device rather than on an assumed overlap.

(ii) **Which observables can be demonstrated, and how do they compare with the same ring running on its native $\chi^{(3)}$?** Section 8 works through sign-tunability, bistability, switching, cross-phase modulation, parametric oscillation, and single-photon anharmonicity.

(iii) **What stops it?** Section 9 quantifies thermal and photorefractive shifts, which are the real competition, and Section 10 turns the answer into a measurement protocol.

Relation to prior work. Cascaded quadratic nonlinearity in $\chi^{(2)}$ microresonators is not new: it is the accepted mechanism behind "quadratic" frequency combs in thin-film LN microresonators [5], and resonantly coupled $\chi^{(2)}$ microrings have been pushed toward single-photon anharmonicity [1, 2], in the lineage of the bulk cascading literature [6, 7, 8]. What is new here is the systematic, observable-by-observable comparison against the *native* $\chi^{(3)}$ of the same ring at matched operating points, the three design rules of Section 6, and the antisymmetric-in-$\Delta\omega$ protocol that separates the cascade from thermal and photorefractive backgrounds.

2 Conventions and the map to the metasurface note

Photon-number convention. For the ring it is cleanest to work in the Hamiltonian (photon-number) convention standard in the $\chi^{(2)}$ microresonator literature. With a the fundamental (FM) mode at ω_1 and b the SH mode at $\omega_2 \approx 2\omega_1$,

$$\hat{H}/\hbar = \omega_1 a^\dagger a + \omega_2 b^\dagger b + g\big[(a^\dagger)^2 b + a^2 b^\dagger\big], \tag{1}$$

which is Eq. (1) of Ref. [1]. Here g is the vacuum photon–photon coupling rate, in rad/s.

Loss rates: two conventions, one factor of two. The microresonator literature uses the HWHM *amplitude* decay rate $\kappa_n = \omega_n/(2Q_n)$; the metasurface note uses the *energy* decay rate $\gamma_n = \omega_n/Q_n$, which is the FWHM linewidth. They differ by exactly a factor of two,

$$\boxed{\gamma_n = 2\kappa_n = \omega_n/Q_n} \tag{2}$$

and confusing them is the single most likely way to get a factor of 4 wrong in R. We quote detunings in units of γ_2 throughout, to stay compatible with the operating-point conventions of the main note ($\Delta\omega = 2\gamma_2$ conservative, $\Delta\omega = \gamma_2$ edge-of-adiabatic). Note $\Delta\omega = 2\gamma_2$ means the SH is detuned by four half-linewidths, i.e. two full linewidths.

Detunings. As in the main note,

$$\Delta\omega \equiv 2\omega_1 - \omega_2, \tag{3}$$

and in a frame rotating at ω_p for a and $2\omega_p$ for b the two detunings are

$$\delta_1 = \omega_1 - \omega_p, \qquad \delta_2 = \omega_2 - 2\omega_p = -\Delta\omega + 2\delta_1. \tag{4}$$

The $2\delta_1$ in (4) matters: scanning the pump across the FM resonance also walks the SH detuning, at twice the rate. All numerics below keep it exactly.

Kerr sign convention. We define every per-photon Kerr coefficient U so that the FM resonance moves as

$$\omega_1^{\text{eff}}(n) = \omega_1 - U\,n, \qquad n = \langle a^\dagger a \rangle, \tag{5}$$

so $U > 0$ is self-focusing (red shift), matching the sign of α_3 in the main note. Native LN Kerr is always self-focusing.

Map to the metasurface note. The note's energy-normalised amplitudes ($|a_n^{\text{note}}|^2$ = stored energy in joules) are $a_n^{\text{note}} = \sqrt{\hbar\omega_n}\,a_n$. Substituting into the note's TCMT equations and matching term by term gives

$$\boxed{g = \beta\sqrt{\hbar\omega_1/2}}, \qquad \beta = \frac{\varepsilon_0\omega_1}{4}\int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{2,i} f_{1,j} f_{1,k}\, d^3r, \tag{6}$$

where β is Eq. (beta) of the main note. The apparent asymmetry between the two equations of motion below (a factor 2 in the FM equation, none in the SH equation) is the same asymmetry that $\omega_2 = 2\omega_1$ hides in the energy normalisation; both forms satisfy Manley–Rowe.

3 Coupled-mode equations and exact elimination of the SH

From (1), adding decay and a single bus waveguide coupled to the FM,

$$\boxed{\frac{da}{dt} = -(i\delta_1 + \kappa_1)a - 2ig\,a^* b + \sqrt{2\kappa_{1,\text{ex}}}\,s_{\text{in}}} \tag{7}$$

$$\boxed{\frac{db}{dt} = -(i\delta_2 + \kappa_2)b - ig\,a^2} \tag{8}$$

with $|s_{\text{in}}|^2$ the input photon flux, $P_{\text{in}} = \hbar\omega_1|s_{\text{in}}|^2$, $\kappa_n = \kappa_{n,0} + \kappa_{n,\text{ex}}$, and $s_{\text{out}} = -s_{\text{in}} + \sqrt{2\kappa_{1,\text{ex}}}\,a$.

The elimination is exact in steady state. Setting $db/dt = 0$ in (8) involves no approximation whatsoever:

$$b_{\text{ss}} = \frac{-ig\,a^2}{i\delta_2 + \kappa_2}. \tag{9}$$

The adiabatic approximation enters only when the FM envelope varies in time. Substituting (9) into (7),

$$\frac{da}{dt} = -\Big[i\delta_1 + \kappa_1 + \frac{2g^2|a|^2}{i\delta_2 + \kappa_2}\Big]a + \sqrt{2\kappa_{1,\text{ex}}}\,s_{\text{in}}, \tag{10}$$

and separating the bracket into real and imaginary parts with $\delta_2 = -\Delta\omega$ gives the two central coefficients of this note:

$$\boxed{U_{\text{casc}}(\Delta\omega) = -\frac{2g^2\,\Delta\omega}{\Delta\omega^2 + \kappa_2^2}} \qquad \text{cascaded Kerr, per photon} \tag{11}$$

$$\boxed{\kappa_{\text{casc}}(\Delta\omega) = \frac{2g^2\,\kappa_2}{\Delta\omega^2 + \kappa_2^2}} \qquad \text{cascaded two-photon loss, per photon} \tag{12}$$

so that the FM sees an effective, intensity-dependent resonance and linewidth

$$\omega_1^{\text{eff}} = \omega_1 - \big[U_{\text{casc}} + U_{\text{nat}}\big]n, \qquad \kappa_1^{\text{eff}} = \kappa_1 + \kappa_{\text{casc}}\,n. \tag{13}$$

Equations (11)–(12) are the microring translation of the main note's $\alpha_3^{\text{eff,casc}} = -|\beta|^2/(\Delta\omega + i\gamma_2/2)$, and reduce to its far-detuned form $U_{\text{casc}} \to -2g^2/\Delta\omega$ for $|\Delta\omega| \gg \kappa_2$ (a 6% correction at $\Delta\omega = 2\gamma_2$).

Sign tunability. $U_{\rm casc}$ is *odd* in $\Delta\omega$: the cascade is self-defocusing for $\Delta\omega > 0$ (SH resonance below $2\omega_1$) and self-focusing for $\Delta\omega < 0$, passing through zero at $\Delta\omega = 0$ where the response is pure loss. $\kappa_{\rm casc}$ is *even*. This parity structure is what makes the measurement of Section 10 work.

Loss-to-Kerr ratio. Dividing (12) by (11),

$$\rho \equiv \frac{\kappa_{\rm casc}}{|U_{\rm casc}|} = \frac{\kappa_2}{|\Delta\omega|} = \frac{\gamma_2}{2|\Delta\omega|}, \tag{14}$$

so $\rho = 1/4$ at the reference point $\Delta\omega = 2\gamma_2$. This single dimensionless number controls every threshold in Section 8; it is the cascaded analogue of the two-photon figure of merit in Kerr materials.

4 Anchoring the absolute scale: g from the measured SHG efficiency

The metasurface sample analysis carried an $O(1)$ unknown ($C_{\rm bridge}$) because the modal overlap had to be inferred from a simulated field integral with an undocumented normalisation. The microring removes this problem, because g is directly measurable.

Solving (7)–(8) in the undepleted-pump limit on double resonance ($\delta_1 = \delta_2 = 0$) gives the intracavity photon numbers $n = 2\kappa_{1,\rm ex}|s_{\rm in}|^2/\kappa_1^2$ and $n_b = 4g^2n^2/\gamma_2^2$, hence an out-coupled SH power $P_{\rm SH} = \hbar\omega_2\, 2\kappa_{2,\rm ex} n_b$. The normalised efficiency $\eta_{\rm SHG} \equiv P_{\rm SH}/P_{\rm in}^2$ is therefore

$$\boxed{\eta_{\rm SHG} = \frac{16\, g^2\, \kappa_{2,\rm ex}\, \kappa_{1,\rm ex}^2}{\hbar\omega_1\, \kappa_2^2\, \kappa_1^4}} \qquad [\mathrm{W}^{-1}]. \tag{15}$$

Check. Equation (15) reproduces Eq. (4) of Ref. [1] exactly. Their expression, at general detuning, is $P_{b,\rm out} = \left[g^2 P_{\rm in}^2/(\hbar\omega_p^2)\right] \omega_b \left[2\kappa_{b1}/(\delta_b^2 + \kappa_b^2)\right] \left[2\kappa_{a1}/(\delta_a^2 + \kappa_a^2)\right]^2$; setting $\delta_a = \delta_b = 0$ and $\omega_b = 2\omega_p$ gives (15) term for term. This is a genuine external check on the whole normalisation chain, and it is the reason the numbers in this note do not inherit the metasurface note's prefactor ambiguity.

What the measured g says about the metasurface note's prefactor. Inserting $|f_n| \sim 1/\sqrt{\varepsilon_0 n_n^2 V_{\rm eff}}$ into (6) gives the first-principles estimate

$$g \;\sim\; \frac{\omega_1 |\chi^{(2)}|}{4\sqrt{\varepsilon_0}\, n_2 n_1^2 \sqrt{V_{\rm eff}}} \sqrt{\frac{\hbar\omega_1}{2}} \;\times\; \frac{2}{\pi}\,\Theta, \tag{16}$$

where $2/\pi$ is the first-order quasi-phase-matching factor for 50%-duty-cycle poling and $\Theta \le 1$ is the transverse overlap. For the device of Ref. [1] ($R = 70\,\mu$m, $A_{\rm eff} \simeq 1\,\mu\mathrm{m}^2$) this gives $g/2\pi = 0.56$ MHz at $\Theta = 1$, a factor 2.1 below the measured 1.2 MHz; on the alternative $\beta = (\varepsilon_0\omega_1/2)M$ branch flagged in §10 of the main note it gives 1.12 MHz.

This does not settle the prefactor question, and we do not claim that it does. A scalar $A_{\rm eff}$ estimate omits the vector-mode normalisation, the tensor contraction, the poling duty cycle and the transverse profile overlap, each an $O(1)$ factor; Ref. [1]'s own full calculation for the same device predicts 1.78 MHz against the 1.2 MHz they fit, so even a careful device-specific calculation carries a 1.5× spread in g (a factor 2.2 in g^2). The observation is recorded only as a direction worth checking: settling it requires a full-wave vector-mode overlap evaluation on a known geometry, not an effective-area estimate. Nothing in this note depends on it, because every microring number below is anchored on the measured g.

Throughout this note we use the *measured* g and do not rely on (16).

Design points. Table 1 lists the three designs used below. Design A is the demonstrated PPLN microring of Ref. [1], the successor of the 250,000 %/W device of Ref. [2]. Design B places the same poling technology on a higher-Q, smaller ring representative of the platform quoted for this project (Q_1^{load} from 10^5 up to 10^6, Q_1^{int} 1–2 $\times 10^6$); g is scaled from A as $V_{\mathrm{eff}}^{-1/2}$. Design C is the same ring *without* poling, relying on modal or cyclic phase matching, for which g is roughly 12× smaller (d_{31} instead of d_{33}, plus a transverse mode-order mismatch), corresponding to $\eta_{\mathrm{SHG}} \sim 10^4$ %/W as reported for mode-phase-matched x-cut LN resonators.

	λ_1	$Q_1^{\mathrm{int}}/Q_1^{\mathrm{load}}$	$Q_2^{\mathrm{int}}/Q_2^{\mathrm{load}}$	R_{ring}	$g/2\pi$
A: PPLN, demonstrated [1]	1.56 μm	$1.8\times10^6/5.4\times10^5$	$5.8\times10^5/4.5\times10^5$	70 μm	1.20 MHz
B: PPLN, target platform	1.55 μm	$2.0\times10^6/1.0\times10^6$	$6.0\times10^5/3.0\times10^5$	50 μm	1.42 MHz
C: no poling (modal/cyclic)	1.55 μm	$2.0\times10^6/1.0\times10^6$	$6.0\times10^5/3.0\times10^5$	50 μm	0.12 MHz

Table 1. Reference designs. Design B is critically coupled at the FM. Material inputs common to all: $n_1 = 2.21$, $n_2 = 2.26$, $n_{g,1} = 2.30$, $n_{g,2} = 2.45$, $A_{\mathrm{eff}} = 1\,\mu\mathrm{m}^2$, $|\chi^{(2)}| = 2d_{33} = 50$ pm/V, and native $n_2^{\mathrm{Kerr}} = 1.8 \times 10^{-19}$ m^2/W (equivalently $\chi^{(3)}_{\mathrm{native}} = 3.1 \times 10^{-21}$ m^2/V^2 through $\chi^{(3)} = \frac{4}{3}\varepsilon_0 c n_1^2 n_2$ in the Boyd convention [10]; this is the conservative, i.e. large, end of the reported LN range). Intrinsic Q's well above those used here have been demonstrated in thin-film LN microrings [3].

For Design B this gives $\kappa_1/2\pi = 96.7$ MHz, $\kappa_2/2\pi = 645$ MHz, $\gamma_2/2\pi = 1.29$ GHz, FSR$/2\pi =$ 415 GHz, $V_{\mathrm{eff}} = 314\,\mu\mathrm{m}^3$, SH photon lifetime $1/\gamma_2 = 123$ ps and FM lifetime $1/\gamma_1 = 823$ ps.

5 The enhancement ratio R

5.1 Native Kerr reference

The comparison architecture is *the same ring* with the SH tuned far away. Its per-photon Kerr shift follows from the round-trip nonlinear phase without any modal-normalisation ambiguity: with $\gamma_{\mathrm{NL}} = \omega_1 n_2^{\mathrm{Kerr}}/(cA_{\mathrm{eff}})$, circulating power $P_{\mathrm{circ}} = U_{\mathrm{cav}} c/(n_{g,1} L)$ and $d\phi/d\omega = L n_{g,1}/c$,

$$\boxed{U_{\mathrm{nat}} = \frac{\hbar\omega_1^2\, c\, n_2^{\mathrm{Kerr}}}{n_{g,1}^2\, V_{\mathrm{eff}}}}, \qquad V_{\mathrm{eff}} = A_{\mathrm{eff}}\, L = 2\pi R_{\mathrm{ring}} A_{\mathrm{eff}}. \tag{17}$$

For Design B, $U_{\mathrm{nat}}/2\pi = 0.805$ Hz per photon. As a sanity check on (17), it predicts a native parametric-oscillation threshold of 9.4 mW for this ring (Section 8.4), consistent with the 10–30 mW Kerr-comb thresholds reported for LN microrings of comparable Q.

5.2 R and its scalings

$$\boxed{R(\Delta\omega) \equiv \frac{|U_{\mathrm{casc}}|}{U_{\mathrm{nat}}} = \frac{2g^2|\Delta\omega|}{(\Delta\omega^2 + \kappa_2^2)\, U_{\mathrm{nat}}}} \tag{18}$$

Two structural facts follow immediately.

R does not depend on the ring size. Both g^2 and U_{nat} scale as $1/V_{\mathrm{eff}}$ — the cascade and the native Kerr are enhanced by mode confinement in exactly the same way — so V_{eff} cancels in (18). Shrinking the ring lowers every threshold power but does not change the cascade's *advantage*. This is the microring counterpart of the main note's observation that R is independent of the FM cavity properties.

$R \propto Q_2$ at fixed $\Delta\omega/\gamma_2$. Writing $\Delta\omega = x\gamma_2 = 2x\kappa_2$,

$$R = \frac{4x\,g^2}{(4x^2+1)\,\kappa_2\,U_{\rm nat}} = \frac{8x\,g^2\,Q_2}{(4x^2+1)\,\omega_2\,U_{\rm nat}}, \qquad Q_2^{\rm cross}\big|_{R=1} = \frac{(4x^2+1)\,\omega_2\,U_{\rm nat}}{8x\,g^2}. \tag{19}$$

At $x = 2$ this gives $Q_2^{\rm cross} = 164$ for the poled designs and 2.4×10^4 for the unpoled one. **With poling, essentially any SH resonance at all beats the native Kerr**; without poling one needs $Q_2 > 2.4 \times 10^4$, which is achievable but leaves only a modest margin.

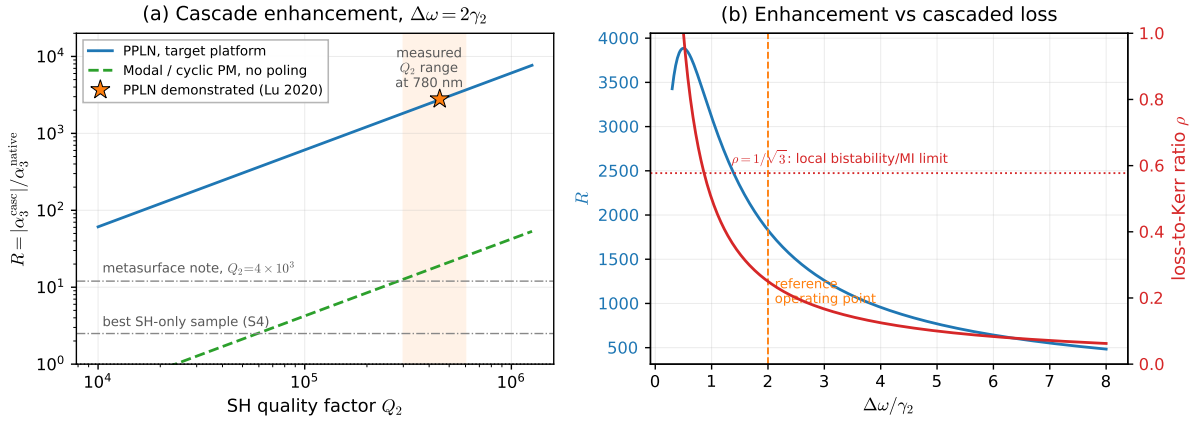


Figure 1. (a) Cascaded enhancement R versus SH quality factor at the reference operating point $\Delta\omega = 2\gamma_2$, for the poled and unpoled designs, with the demonstrated device marked. The horizontal reference lines are the metasurface note's $R \simeq 12$ at $Q_2 = 4 \times 10^3$ and the best fabricated SH-only sample ($R^{(1\mathrm{R})} = 2.5$). (b) The operating-point trade-off: R falls as $1/\Delta\omega$ while the cascaded loss-to-Kerr ratio ρ falls at the same rate, and bistability is lost above $\rho = 1/\sqrt{3}$.

5.3 Numbers

design	$g/2\pi$ [MHz]	Q_2	R	$U_{\rm casc}/2\pi$ [Hz/ph]	$P_{\rm bist}$ [μW]	$P_{\rm MI}$ [μW]
A: PPLN demonstrated	1.20	4.5×10^5	2796	-1587	72.8	20.3
B: PPLN target	1.42	3.0×10^5	1828	-1472	36.6	8.4
C: no poling	0.12	3.0×10^5	13	-10	7039	1216
native Kerr, same ring	—	—	1	$+0.80$	14405	9357

Table 2. Cascaded response at cold-cavity $\Delta\omega = 2\gamma_2$. The bistability column is the exact threshold, minimised over pump detuning with the cascaded coefficients evaluated at the pump-frame mismatch $\Delta_{\rm eff} = \Delta\omega - 2\delta_1$ (Section 8.2); the parametric column is the multimode Bogoliubov threshold of Section 8.4 at $D_2/2\pi = 10$ MHz. The native row uses the Design B ring with the SH detuned away; its parametric entry is the standard single-sideband-pair Kerr estimate.

The SH coupling condition is a free 2×. Table 1 takes $Q_2^{\rm load} = 3 \times 10^5$ with $Q_2^{\rm int} = 6 \times 10^5$, i.e. a critically coupled SH. The cascade does not need the SH extracted — it needs it *stored* — so leaving the SH port under-coupled raises $Q_2^{\rm load}$ toward $Q_2^{\rm int}$ and R with it, to 3.7×10^3 in the fully under-coupled limit. The price is that the SH power is then dissipated internally rather than carried off chip, which doubles the cascade's own heat load (Section 9), and that g can no longer be calibrated on that device. The headline numbers below keep the conservative critically coupled choice.

For Design B the equivalent material parameters are

$$n_2^{\rm eff} = -3.3 \times 10^{-16}\ \mathrm{m^2/W}, \qquad \chi^{(3)}_{\rm eff} = -5.7 \times 10^{-18}\ \mathrm{m^2/V^2}, \tag{20}$$

i.e. a self-defocusing nonlinearity three orders of magnitude above native LN and comparable to the largest reported non-resonant material Kerr coefficients — but, unlike those, sign-tunable and available in a low-loss, TPA-free host at telecom.

Consistency with the metasurface note. Inverting the main note's boxed result $\chi^{(3)}_{\rm casc} = -\omega_1|\chi^{(2)}|^2\eta_{\rm geom}/(6n_2^2\Delta\omega)$ with the numbers above returns $\eta_{\rm geom} = 0.93$ for Design B and 0.006 for Design C. The unpoled value is squarely in the range found for the fabricated metasurface samples (0.01–0.04); the poled value sits at the physical ceiling $\eta_{\rm geom} \leq 1$, which is again the prefactor question of Section 4 (on the larger-prefactor branch it becomes $\simeq 0.23$, a comfortable value for quasi-phase-matched interaction). Either way the ordering is the message: **periodic poling over a full circumference is what a metasurface meta-atom cannot do**, and it is worth a factor of ~ 100 in R.

6 Design rules specific to the ring

6.1 Rule 1: the cascade costs no bandwidth here

In the metasurface, $\gamma_2 < \gamma_1$ and the useful bandwidth was $\min(\gamma_1, \gamma_2) = \gamma_2$: the cascade was the bottleneck, and the sum rule of note §7 bit immediately. In a ring,

$$\frac{\gamma_2}{\gamma_1} = \frac{\omega_2/Q_2}{\omega_1/Q_1} = \frac{2Q_1}{Q_2} \;\simeq\; 6.7 \quad \text{(Design B)}, \tag{21}$$

so the SH is far broader than the FM and the FM cavity remains the bottleneck, exactly as it would be for a native-Kerr ring.

A susceptibility argument makes the scale plausible — a modulation of $|a|^2$ at Ω drives the SH through $[i(\delta_2 - \Omega) + \kappa_2]^{-1}$, so the cascaded coefficient is obtained from (11) by $\Delta\omega \to \Delta\omega + \Omega$, rolling off at $|\Delta\omega|/2\pi = 2.6$ GHz against $\kappa_1/2\pi = 97$ MHz — but a susceptibility scale is not a transfer function. We therefore computed the actual small-signal response: linearise the driven system (7)–(8) about its CW fixed point and solve $(-i\Omega\mathbb{1} - J)\,\delta y = \hat{p}$ for a modulation of the input power, with the drive perturbation taken along p (not along the real axis — with a gauged real, $p = \sqrt{n}[(\kappa_1 + \kappa_{\rm casc}n) + i(\delta_1 - Un)]$). The reference is a *matched* single-mode cavity: same photon number, same total per-photon Kerr $U_{\rm nat} + U_{\rm casc}$, same nonlinear loss $\kappa_{\rm casc}$, but responding instantaneously. Any difference is then attributable solely to the SH's finite response time.

At 20 μW and $\Delta\omega = 2\gamma_2$ the cascade's 3 dB bandwidth is 421 MHz against 454 MHz for the matched instantaneous reference: **the SH costs 7% of modulation bandwidth**, and 2% at 2 μW. Equivalently, the four eigenvalues of the full Jacobian separate cleanly into an FM-like pair at $|\lambda| = 2.45\,\kappa_1$ and an SH-like pair at $|\lambda| = 29.6\,\kappa_1$, a factor 12 apart. The resonator, not the cascade, sets the speed. Figure 8(b) shows both curves.

6.2 Rule 2: locality of the cascaded response

The exact elimination of the SH in the time domain is

$$b(t) = -ig\int_0^\infty e^{-(\kappa_2 + i\delta_2)\tau}\, a^2(t-\tau)\, d\tau, \tag{22}$$

so the memory kernel decays at the rate κ_2 but *oscillates* at δ_2. What controls whether the response may be treated as instantaneous is the modulus

$$\boxed{\Omega_{\rm env} \;\ll\; \sqrt{\kappa_2^2 + \delta_2^2}} \tag{23}$$

where $\Omega_{\rm env}$ is the bandwidth of the FM envelope, not κ_2 alone. Detuning helps: at the reference operating point $\delta_2 = -2\gamma_2 = -4\kappa_2$, so the right-hand side is $\sqrt{17}\,\kappa_2 = 27.5\,\kappa_1$, four times more forgiving than the naive $\kappa_2/\kappa_1 = 6.7$.

A crude reading of (23) with $\Omega_{\rm env} \sim \kappa_1$ and $\delta_2 \to 0$ would give $Q_2 \lesssim 2Q_1$; we quote (23) instead, because that inequality is not a general boundary and is unnecessarily pessimistic wherever the device is actually operated. The measured consequence is the 7% bandwidth cost of Section 6.1. What does remain true is the qualitative trade: $R \propto Q_2$ wants a narrow SH resonance, locality wants a broad one, and the two meet near $Q_2 \sim Q_1$. Far beyond that the dynamics ceases to be Kerr-like and one is in the quadratic-soliton regime.

6.3 Rule 3: the local loss-to-Kerr limit $\rho < 1/\sqrt{3}$

Using (13), the CW steady state satisfies

$$|p|^2 = n\Big[(\kappa_1 + \kappa_{\rm casc} n)^2 + (\delta_1 - Un)^2\Big], \qquad |p|^2 = \frac{2\kappa_{1,\rm ex} P_{\rm in}}{\hbar\omega_1}, \tag{24}$$

a cubic in n. Turning points require the discriminant of $dP/dn = 3(U^2 + \kappa_{\rm casc}^2)n^2 + 4(\kappa_1\kappa_{\rm casc} - \delta_1 U)n + (\delta_1^2 + \kappa_1^2)$ to be positive, i.e.

$$4(\rho - s)^2 > 3(1+\rho^2)(s^2+1), \qquad s \equiv |\delta_1|/\kappa_1, \tag{25}$$

with the pump on the side the resonance moves toward, $\operatorname{sign}\delta_1 = \operatorname{sign} U$. As $s \to \infty$ this requires $4 > 3(1+\rho^2)$, i.e.

$$\boxed{\rho < \frac{1}{\sqrt{3}} \simeq 0.577 \quad \Longleftrightarrow \quad |\Delta_{\rm eff}| > \frac{\sqrt{3}}{2}\,\gamma_2 \simeq 0.87\,\gamma_2} \tag{26}$$

Equation (25) also raises the required pump detuning: $|\delta_1|/\kappa_1$ grows from $\sqrt{3} = 1.73$ at $\rho = 0$ to 3.50 at $\rho = 1/4$ and diverges as $\rho \to 1/\sqrt{3}$. This is the cascaded analogue of the requirement that the two-photon figure of merit be small enough for Kerr bistability, and the same inequality reappears in Section 8.4 as the single-sideband-pair parametric condition.

It is a condition on $\Delta_{\rm eff}$, not on the cold cavity. This distinction is easy to get wrong and we got it wrong in an earlier draft. The coefficients U and $\kappa_{\rm casc}$ must be evaluated at the *pump-frame* mismatch $\Delta_{\rm eff} = \Delta\omega - 2\delta_1$ of (4), which moves as the pump is scanned. Freezing them at the cold-cavity $\Delta\omega$ turns (26) into an apparent fabrication requirement $|\Delta\omega| > 0.87\gamma_2$, and that is **false**: the experimenter chooses δ_1, and a pump detuned far enough to one side supplies whatever $\Delta_{\rm eff}$ is needed. Solving the threshold problem exactly — minimising the fold power over δ_1 with $U(\Delta_{\rm eff})$ and $\kappa_{\rm casc}(\Delta_{\rm eff})$ — gives a threshold that is finite for *every* cold-cavity detuning, including $\Delta\omega = 0$, where it is 304 μW. The frozen model diverges there (Figure 5(b)). What survives of Rule 3 is a genuine local statement: at the operating point, the pump-frame loss-to-Kerr ratio must satisfy $\rho < 1/\sqrt{3}$.

7 Branch stability and the dispersive-breakdown scale

The main note's saturation criterion $|\beta a_1| \to |\Delta\omega|$ translates, via (6), into $\sqrt{2}\,g\sqrt{n} = |\Delta\omega|$, i.e.

$$n_{\rm sat} = \frac{\Delta\omega^2}{2g^2}, \qquad \text{whence} \qquad |U_{\rm casc}|\, n_{\rm sat} \simeq |\Delta\omega|. \tag{27}$$

The cascaded resonance shift reaches order $\Delta\omega$ itself at this scale, which is $4\kappa_2/\kappa_1 = 8Q_1/Q_2$ FM linewidths. It should be read as the point at which the coherent SH coupling stops being a small perturbation on the detuning — a dispersive-breakdown scale — not as an exact saturation of the equations of motion. The steady-state elimination (9) remains algebraically exact at any amplitude; what fails beyond this scale is the interpretation of the result as a weak, local Kerr nonlinearity.

We checked (27) against a direct linear-stability analysis: linearising (7)–(8) about the CW fixed point (with a real by gauge, b from (9)) gives a 4×4 real Jacobian whose largest eigenvalue real part crosses zero at a critical photon number n_c. Over $\Delta\omega/\gamma_2 \in [0.6, 8]$ the numerical n_c exceeds n_{sat} by a factor 1.13–1.92 (median 1.14), so (27) is a mildly conservative estimate and the scaling is correct. Figure 2 shows both.

For Design B at $\Delta\omega = 2\gamma_2$ this is $n_c = 2.0 \times 10^6$ photons, 153 μW of input power, and a resonance shift of $30\,\kappa_1$ — an order of magnitude above the $\sqrt{3}\,\kappa_1$ needed for bistability and the $1\,\kappa_1$ needed for switching. Two qualifications: the scan is along the $\delta_1 = 0$ path and the Jacobian used here omits the native Kerr (which is 10^{-3} of the cascaded term), and the loss of stability of *this* branch does not imply that no CW state exists. With those caveats, **dynamic range is not the limiting factor**; thermal and photorefractive effects (Section 9) reach that territory first.

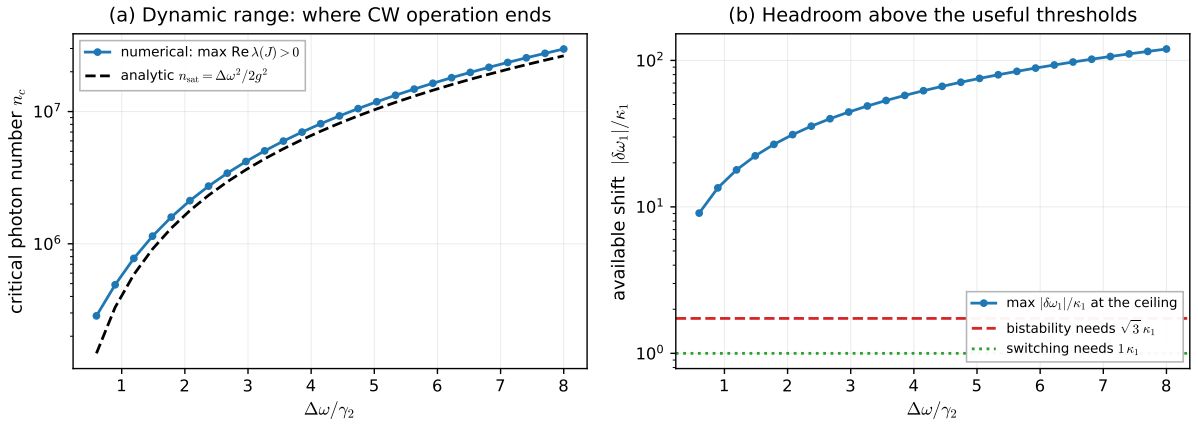


Figure 2. (a) Critical photon number from the 4×4 Jacobian compared with the analytic ceiling $n_{\mathrm{sat}} = \Delta\omega^2/2g^2$. (b) The resonance shift available at that ceiling, in FM linewidths, against the shifts required for bistability and for switching.

8 Observables

8.1 Sign-tunable Kerr: the primary signature

The cleanest demonstration is not a large nonlinearity — large nonlinearities can be mimicked by thermal and photorefractive effects — but a nonlinearity that *changes sign* under a control the competing mechanisms do not respond to. From (11), walking the SH resonance through $2\omega_1$ takes the FM resonance shift from red ($\Delta\omega < 0$, self-focusing) through zero to blue ($\Delta\omega > 0$, self-defocusing), with extrema at $\Delta\omega = \mp\kappa_2$ of magnitude g^2/κ_2. Native LN Kerr, thermal shifts and photorefractive shifts are all one-signed.

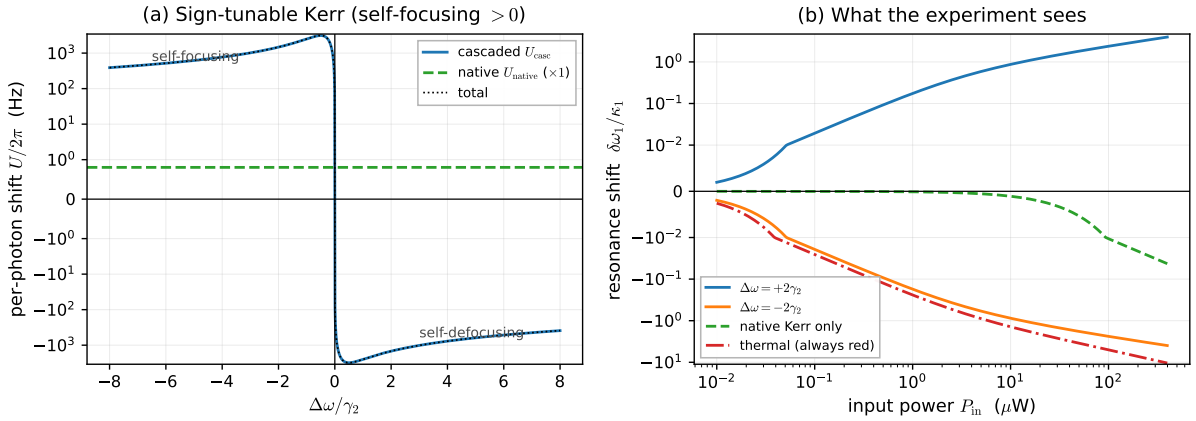


Figure 3. (a) Per-photon Kerr coefficient versus SH detuning (Design B): the cascade is odd in $\Delta\omega$ and dwarfs the native contribution over the whole useful range. (b) The corresponding FM resonance shift versus input power, with n computed self-consistently (including the cascaded loss and self-detuning), for both signs of $\Delta\omega$, against the native-Kerr and thermal backgrounds. A blue shift is a strong indicator of the cascade, but not by itself proof: photorefraction in thin-film LN can also shift a resonance blue (Section 9). The decisive test is the controlled reversal under $\Delta\omega \to -\Delta\omega$.

How to sweep $\Delta\omega$. Two knobs, both native to LN:

- **Temperature.** Each resonance tunes fast ($d\omega_1/dT \simeq -2\pi \times 2.8$ GHz/K and $d\omega_2/dT \simeq -2\pi \times 5$–$7$ GHz/K), but $\Delta\omega = 2\omega_1 - \omega_2$ is a *difference* of two nearly equal quantities and therefore tunes slowly. Its rate depends on the dispersion of dn/dT between 1.55 μm and 775 nm, which is not well pinned: taking $(dn/dT)_{\rm SH} = 3.3$–4.5×10^{-5} K^{-1} against 3.3×10^{-5} at the FM gives $d\Delta\omega/dT$ anywhere from -0.34 to $+1.6$ GHz/K. *Both the sign and the magnitude must be measured*, but in every case the scale is $\sim \gamma_2$ per few kelvin: sweeping $\Delta\omega$ over $\pm 2\gamma_2$ takes 2–9 K. The near-cancellation is a gift — $\Delta\omega$ is a coarse-grained, easily stabilised coordinate even though each resonance individually is not.
- **Electro-optic (the important one).** With $r_{33} = 30.8$ pm/V at the FM and 34 pm/V at the SH, the same difference structure gives $d\Delta\omega/dE = -2\pi \times 3.0$ GHz per V/μm: already $2.4\,\gamma_2$ at 1 V/μm, $4.7\,\gamma_2$ at 2 V/μm and $23\,\gamma_2$ at 10 V/μm. Modulator-level drive voltages therefore sweep $\Delta\omega$ across several SH linewidths, and the Pockels effect supports tens of GHz of modulation bandwidth in this platform, so $\Delta\omega$ can be dithered far faster than any thermal or photorefractive response. This is the single most useful thing lithium niobate contributes beyond its $\chi^{(2)}$.

8.2 Bistability and threshold power

Solving (24) exactly — minimising the fold power over δ_1 with the coefficients taken at $\Delta_{\rm eff} = \Delta\omega - 2\delta_1$, as Section 6.3 requires — gives the hysteresis loops of Figure 4. For Design B the threshold falls from 14.4 mW (native, where there is no $\Delta_{\rm eff}$ dependence and the closed form is exact) to 36.6 μW at cold $\Delta\omega = 2\gamma_2$, a factor 394. Note this is *less* than $R = 1828$: the cascaded two-photon loss costs roughly a factor 5, both by forcing a larger pump detuning (Rule 3) and by broadening the resonance. The fixed-coefficient treatment overestimates the threshold by 26% here (46 μW) and diverges as $\Delta\omega \to \frac{\sqrt{3}}{2}\gamma_2$, where the exact answer stays finite.

It is worth being blunt about what the native column of Table 2 means. At the 14.4 mW required for native Kerr bistability, the lumped thermal model of Section 9 predicts a thermal shift of $\sim 1.4 \times 10^3$ linewidths. **Native Kerr bistability is not merely harder in this ring, it is thermally inaccessible by three orders of magnitude**; the cascade does not just

improve the number, it is what makes the observable exist at all. Optimising over the operating point (Figure 5(b)) gives

$$\Delta\omega^{\mathrm{opt}} \simeq 3.3\,\gamma_2, \qquad P_{\mathrm{bist}} = 31.7\,\mu\mathrm{W}, \qquad \text{at } \delta_1 = -2.4\,\kappa_1, \quad \Delta_{\mathrm{eff}} = 3.7\,\gamma_2, \tag{28}$$

a shallow optimum arising because $R \propto 1/\Delta\omega$ pushes one way while $\rho \propto 1/\Delta\omega$ pushes the other. (The frozen model's apparent optimum of $3.7\gamma_2$ was in fact the *pump-frame* mismatch at the true optimum, which is why it looked numerically reasonable.) We therefore recommend a cold-cavity $\Delta\omega \simeq 3\gamma_2$ rather than the metasurface note's $2\gamma_2$ for any threshold-limited experiment.

The transmission lineshape is the direct observable: Figure 4(b) shows the resonance tilting to opposite sides for $\Delta\omega = \pm 2\gamma_2$ at $30\,\mu$W. *The tilt reversal under a sign change of $\Delta\omega$, at fixed power, is the measurement.*

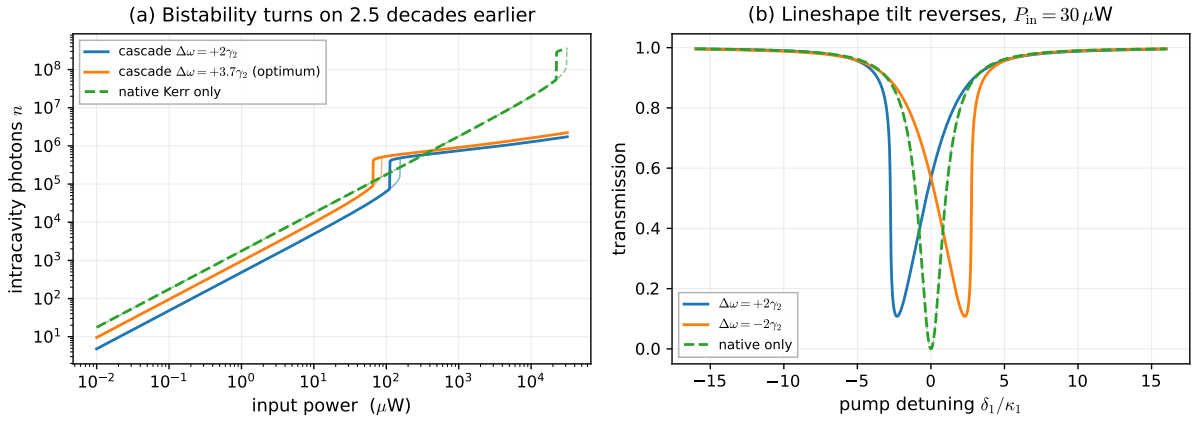


Figure 4. (a) Steady-state intracavity photon number versus input power for the cascade at two operating points and for the native Kerr, each at 1.45× its own minimum bistable detuning. The $-\Delta\omega$ branches give identical $|n|$ curves and are not drawn. (b) Transmission lineshapes at $30\,\mu$W: the tilt reverses with the sign of $\Delta\omega$ while the native response stays nearly symmetric.

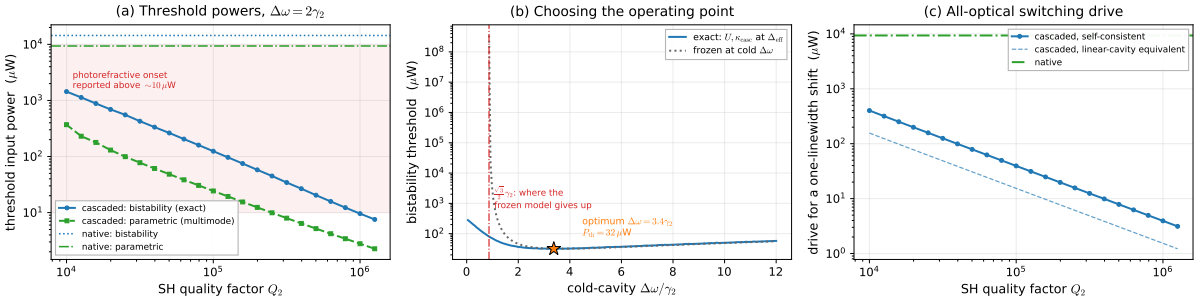


Figure 5. (a) Bistability and parametric thresholds versus Q_2, cascaded and native, with the photorefractive onset reported in Ref. [1] shaded. (b) Threshold versus operating point, showing the shallow optimum near $\Delta\omega \simeq 3.7\gamma_2$ and the divergence at $\Delta\omega = \frac{\sqrt{3}}{2}\gamma_2$; the dotted curve is the idealisation with the cascaded loss switched off, whose gap to the solid curve is the price of ρ. (c) Drive power and intracavity energy for a one-linewidth shift.

8.3 Self- and cross-phase modulation, and switching

The per-photon shift (11) is self-phase modulation. For cross-phase modulation between two FM modes a_1, a_2, the mediating SH mode is the one at the *sum* azimuthal number, and the interaction Hamiltonian carries the usual non-degenerate factor, $\hat{H}/\hbar \supset 2g(a_1^\dagger a_2^\dagger c + \mathrm{h.c.})$. Eliminating c

gives an XPM coefficient

$$U_{\mathrm{XPM}} = -\frac{4g^2\Delta\omega_c}{\Delta\omega_c^2 + \kappa_2^2} = 2\,U_{\mathrm{casc}}(\Delta\omega_c), \tag{29}$$

i.e. the familiar factor-of-two XPM/SPM ratio of an ordinary Kerr medium, but evaluated at the detuning $\Delta\omega_c$ of *that* SH mode, which is generally different (Section 8.4). The factor is exactly two only if the two triplets share the same three-wave overlap, $g_{12c} = g_{11b}$; in a real ring they differ through the frequencies, transverse profiles, tensor projections and QPM overlap, so read it as "two, up to the overlap ratio".

For switching, the relevant figure is the drive needed for a one-linewidth shift, $n_{\mathrm{sw}} = \kappa_1/|U|$:

	cascaded (B, $\Delta\omega = 2\gamma_2$)	native
photons for $\delta\omega_1 = \kappa_1$	6.6×10^4	1.2×10^8
intracavity switching energy	8.4 fJ	15.4 pJ
linear-cavity-equivalent loading	5.1 μW	9.4 mW
self-consistent CW drive, $\delta_1 = 0$	13.1 μW	18.7 mW
self-consistent, optimal retuning	7.6 μW	—

The distinction in the last three rows matters. The intracavity energy is the clean figure of merit. The "linear-cavity-equivalent" row inverts the *linear* loading formula and is what a naive estimate returns; it understates the real drive by 2.6×, because at the switching point the resonance has already moved by a full linewidth and the cavity is no longer being driven on resonance. The self-consistent rows solve (24) for the power that actually reaches n_{sw}. The cascade brings all-optical switching in LN from the "needs an amplifier" regime into the "needs a laser diode" regime, at a switching energy competitive with the best resonant Kerr switches, and with a switching *speed* limited by κ_1 rather than by the nonlinearity (Rule 1).

8.4 Parametric oscillation and frequency combs

The degenerate four-wave-mixing process $2a_p \to a_s + a_i$ with $m_s + m_i = 2m_p$ is mediated by the *same* SH mode that produces the pump's SPM, so it inherits the full cascaded coupling. A gain-equals-loss estimate is not adequate here, so we solve the linear-stability problem properly.

Multimode Bogoliubov threshold. Retain the pump a_0, a sideband pair $a_\pm$ at $m_p \pm q$, the SH mode b_0 at $2m_p$ driven by the pump, and the SH sidebands $b_\pm$ at $2m_p \pm q$:

$$\hat{H}/\hbar \supset g\big(a_0^{\dagger 2} b_0\big) + 2g\big(a_+^\dagger a_-^\dagger b_0\big) + 2g\big(a_0^\dagger a_+^\dagger b_+\big) + 2g\big(a_0^\dagger a_-^\dagger b_-\big) + \mathrm{h.c.} \tag{30}$$

Linearising about the CW pump closes a 4×4 system in $(\delta a_+, \delta a_-^*, \delta b_+, \delta b_-^*)$ whose largest eigenvalue real part gives the parametric gain; the threshold is the lowest input power at which it turns positive, minimised over δ_1 and q. The parametric gain enters through b_0, while the pump-induced cross-phase shift and cascaded loss on the sidebands enter through $b_\pm$, whose detunings walk off with q because the FM and SH free spectral ranges differ. (This assumes equal modal overlaps for every triplet, and neglects the native Kerr on the sidebands, which is $\sim 10^{-3}$ of the cascaded term.)

For Design B this gives $P_{\mathrm{MI}} = 8.4\,\mu$W against 9.4 mW natively, a factor 1100. The result is insensitive to the ring dispersion — 8.1, 8.1 and 8.4 μW for $D_2/2\pi$ = 1, 10 and 50 MHz — because the instability simply selects whichever q brings the sideband into resonance. For reference, the single-mode gain-equals-loss estimate gives 6.8 μW and its self-consistent version 12.1 μW, so the heuristics bracket the answer to within a factor of 1.5 but neither is reliable on its own.

The local single-pair condition. Adiabatically eliminating $b_\pm$ when they are degenerate with b_0 (no walk-off) reduces the 4×4 to $\left[\begin{smallmatrix} A & B \\ B^* & A^* \end{smallmatrix}\right]$ with $A=-(i\delta_q+\kappa_1)+2iUn-2\kappa_{\mathrm{casc}}n$ and $B=iUn-\kappa_{\mathrm{casc}}n$, whose eigenvalues are $\mathrm{Re}\,A\pm\sqrt{|B|^2-(\mathrm{Im}\,A)^2}$. Instability at the optimal sideband detuning $\delta_q=2Un$ requires $n|U|\sqrt{1+\rho^2}>\kappa_1+2\rho|U|n$, i.e.

$$n_{\mathrm{MI}}=\frac{\kappa_1}{|U|\left(\sqrt{1+\rho^2}-2\rho\right)},\qquad \text{possible only for}\quad \rho<\frac{1}{\sqrt{3}}, \tag{31}$$

the same $\rho<1/\sqrt{3}$ *that governs bistability* (Section 6.3). The 4×4 reproduces (31) to 9×10^{-4} in the zero-walk-off limit, which is how we validated it. With walk-off the SH sidebands are pushed off resonance, the cross-phase term is suppressed, and the constraint on ρ relaxes — one reason the full calculation is needed.

Two ring-specific consequences:

An extra degree of freedom for dispersion engineering. Modulation instability of the bright-soliton kind requires $\beta_2\gamma_{\mathrm{NL}}<0$, and native LN is self-focusing. It would be too strong to say that combs therefore *demand* anomalous dispersion — normal-dispersion microresonators support coherent dark-pulse (platicon) states, and those are routinely generated [9]. The defensible claim is narrower: a sign-tunable $\chi^{(3)}$ adds a degree of freedom to MI and phase-matching engineering that a fixed-sign Kerr medium does not have, and in particular lets the *same* ring be operated in either MI regime by moving $\Delta\omega$. A related capability is nulling: at the $\Delta\omega$ where $U_{\mathrm{casc}}=-U_{\mathrm{nat}}$ the total Kerr vanishes, giving a Kerr-free high-Q LN ring.

But the cascade is not a uniform $\chi^{(3)}$ across the comb. Sum-frequency generation of FM lines (m_p+q_1, m_p+q_2) lands on the SH mode $2m_p+j$ with $j=q_1+q_2$, whose detuning walks off as

$$\Delta\omega_j=\Delta\omega-j(\Omega_2-\Omega_1),\qquad \Omega_2-\Omega_1=\Omega_1\Big(\frac{n_{g,1}}{n_{g,2}}-1\Big)=-2\pi\times25.4\ \mathrm{GHz} \tag{32}$$

for Design B, which is $19.7\,\gamma_2$ *per mode index*. Only $j=0$ — which covers the pump SPM and the degenerate parametric process above — is resonant; $j=\pm1$ is already suppressed tenfold (Figure 6(b)). So the parametric *threshold* reduction is rigorous, but the subsequent comb dynamics is not Lugiato–Lefever-like, because XPM among comb lines is strongly mode-selective. This is the same physics that distinguishes quadratic combs from Kerr combs in the literature [5]. Making the cascade broadband requires group-index matching: keeping $|\Delta\omega_j|$ within a factor of a few of $|\Delta\omega|$ out to $|j|=J$ needs $|\Delta n_g|/n_g\lesssim\Delta\omega/(J\,\Omega_1)$, which is 6×10^{-4} for $J=10$ at this operating point and scales as $1/J$. The number is therefore tied to a chosen comb span, not universal. It is a demanding but well-posed dispersion-engineering target and, in our view, the most interesting design question this analysis raises.

8.5 Single-photon anharmonicity: the cascade is the wrong regime

For photon blockade one needs a per-photon anharmonicity comparable to or exceeding the linewidth (the precise condition depends on the dressed-state structure and the linewidths, so the "$=1$" line in Figure 6(a) is schematic). In the dispersive (cascaded) regime, with $\Delta\omega\gtrsim\gamma_2=2\kappa_2$,

$$\frac{|U_{\mathrm{casc}}|}{\kappa_1}=\frac{2g^2}{|\Delta\omega|\,\kappa_1}\lesssim\frac{g}{\kappa_1}\cdot\frac{g}{\kappa_2}, \tag{33}$$

the *product* of two small numbers, whereas resonant operation ($\Delta\omega\to0$, no Kerr description) has the single small number g/κ_1 as its figure of merit. For Design B, $|U_{\mathrm{casc}}|/\kappa_1=1.5\times10^{-5}$ against

$g/\kappa_1 = 1.5 \times 10^{-2}$. **Cascading buys three orders of magnitude of classical nonlinearity and loses three orders of magnitude of quantum anharmonicity**, and the two statements are the same statement: the cascade converts a strong resonant coupling into a weak dispersive one in exchange for a Kerr-like, classically useful response. Quantum work on this platform should stay resonant — which is what Ref. [1] in fact pursues: it is a demonstration of resonant $\chi^{(2)}$ single-photon anharmonicity (and a proposal for unconventional blockade), not of the dispersive cascade analysed here. We use it only as the source of a measured g.

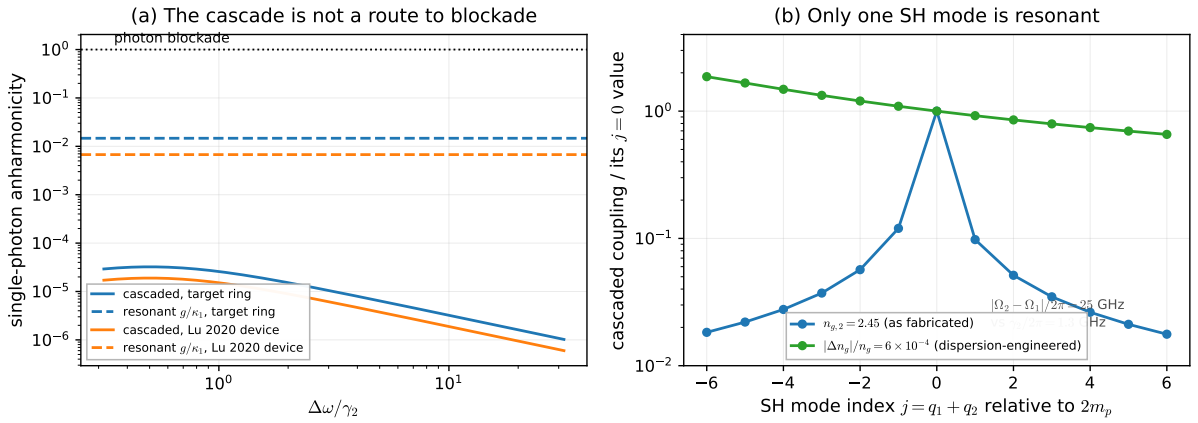


Figure 6. (a) Dispersive versus resonant single-photon figure of merit. (b) Cascaded coupling for the SH mode $2m_p + j$ relative to $j = 0$: the FM/SH free-spectral-range walk-off detunes neighbouring SH modes by $\sim 20\gamma_2$ per index, so only $j = 0$ participates unless the group indices are matched.

9 What actually competes: thermal and photorefractive shifts

At the powers where the cascade produces a linewidth-scale shift ($\sim 10\,\mu$W), two other mechanisms produce shifts of the same order.

Thermal. With $dn/dT = 3.3 \times 10^{-5}$ K^{-1} and a thermal resistance $R_{\rm th} \sim 3 \times 10^4$ K/W, the FM resonance red-shifts by $|d\omega_1/dT|\, R_{\rm th} P_{\rm abs}$. Getting $P_{\rm abs}$ right requires care about bookkeeping. The power leaving the FM through the SH channel is $P_{\rm casc} = 2\hbar\omega_1\kappa_{\rm casc} n^2$, but this is not all local heat: only the fraction $\kappa_{2,0}/\kappa_2$ stays on chip (the rest exits the SH port), and of that only the absorptive part becomes heat rather than scattered light. We therefore take

$$P_{\rm abs} = f_{\rm abs}\Big[\underbrace{2\hbar\omega_1\kappa_{1,0} n}_{\text{FM intrinsic}} + \frac{\kappa_{2,0}}{\kappa_2}\,\underbrace{2\hbar\omega_1\kappa_{\rm casc} n^2}_{\text{SH channel}}\Big]. \tag{34}$$

$P_{\rm casc}$ is even in $\Delta\omega$ and *peaks* at $\Delta\omega = 0$, where $\kappa_{\rm casc} \to 2g^2/\kappa_2$; it does not diverge, and with n solved self-consistently it is bounded by the input power.

For Design B at $20\,\mu$W the FM intrinsic loss carries $6.3\,\mu$W and the SH channel $3.8\,\mu$W (half of it out the port), giving a thermal shift of $2.1\,\kappa_1$ at $f_{\rm abs} = 0.3$ and $0.35\,\kappa_1$ at $f_{\rm abs} = 0.05$, against a cascaded Kerr shift of $1.2\,\kappa_1$. **Thermal effects are therefore within a factor of two of the cascade in either direction** — not negligible, and not necessarily dominant either. The earlier version of this estimate counted the whole SH channel as local heat and overstated the thermal shift by 2.3×. The magnitude should be read as illustrative until the absorption partition is measured; the conclusion that it is comparable to the cascade is robust across the whole plausible range (Figure 7(a)).

Photorefractive. Reference [1] reports the SHG efficiency of their PPLN microring degrading above $\sim 10\,\mu$W of on-chip pump, attributed to a photorefractive shift of the FM/SH frequency mismatch — i.e. precisely to an uncontrolled drift of $\Delta\omega$. This sits exactly at the powers we want. Mitigations are known: removing the dielectric cladding largely suppresses the effect in LN microrings [4]; operating at elevated temperature and using MgO-doped films are the standard alternatives. This should be treated as the principal experimental risk, and as a reason to run the experiment with $\Delta\omega$ actively stabilised rather than set and left.

Why the measurement still works. To leading order the two backgrounds are *even* in $\Delta\omega$ — thermal loading depends on absorbed power, which depends on $\kappa_{\mathrm{casc}} \propto 1/(\Delta\omega^2 + \kappa_2^2)$; photorefractive charge generation depends on the SH intensity, likewise even — while the cascaded Kerr is *odd*. Antisymmetrising the measured shift,

$$\boxed{\delta\omega_1^{\mathrm{casc}}(\Delta\omega) = \tfrac{1}{2}\left[\delta\omega_1^{\mathrm{meas}}(\Delta\omega) - \delta\omega_1^{\mathrm{meas}}(-\Delta\omega)\right]} \tag{35}$$

cancels them to the extent that they are even. Within the model we verified that (35) recovers $U_{\mathrm{casc}}n$ from a thermally dominated total with a relative error of 2×10^{-3} — the residual being the genuine even-order cross term between the native and cascaded Kerr (Figure 7(b)).

We should be clear about what that does and does not establish. It shows the arithmetic of the decomposition is sound for an *even* background. It does not establish that photorefraction is exactly even: photorefractive charge generation, transport, trapping and the resulting space-charge field depend on crystal orientation, surfaces, cladding and history, and no photorefractive dynamical model is implemented here. The honest claim is that the antisymmetric component is a strong signature of the cascade and suppresses backgrounds that are even under a controlled reversal of Δ_{eff}, and that any residual odd background must be bounded experimentally — for instance by repeating the measurement at two pump powers, since the cascade is strictly linear in n while photorefractive saturation is not.

Two conditions the protocol must respect. The parity argument is exact only under them.

(i) The pump must sit at $\delta_1 = 0$, or the readout must be a weak probe. At $\delta_1 \neq 0$ the steady-state cubic (24) contains $(\delta_1 - Un)^2$, whose cross term $-2\delta_1 Un$ is *odd* in U and hence in $\Delta\omega$: the intracavity photon number itself then differs between $\pm\Delta\omega$ and the measured shift is no longer a clean odd function. At $\delta_1 = 0$ the cubic depends on U only through U^2, so n is even and the extraction is exact — which is why Figure 7(b) is computed there. This is not a fine point: repeating the numerical extraction at finite pump detuning gives a relative error of 0.3% at $\delta_1 = 0$, but 130% at $\delta_1 = 0.5\,\kappa_1$ and 340% at $\delta_1 = 2\,\kappa_1$. The method fails outright if the pump is allowed to sit off resonance. In practice either lock the pump to the (thermally shifted) resonance, or use a strong fixed-frequency pump with a weak counter-propagating probe to read the resonance position; the low-power limit, where n is independent of U altogether, also works.

(ii) The zero of $\Delta\omega$ must be calibrated in the same run. The photorefractive effect acts *on* $\Delta\omega$ *itself*, so a slow space-charge field displaces the point about which one antisymmetrises. Under fast dithering it cannot follow and acts as a static offset $\Delta\omega \to \Delta\omega + \Delta\omega_{\mathrm{PR}}$, which converts a fraction of the odd signal into an even one and vice versa. Anchoring $\Delta\omega = 0$ to the maximum of the generated SH power, remeasured alongside the dither, removes it.

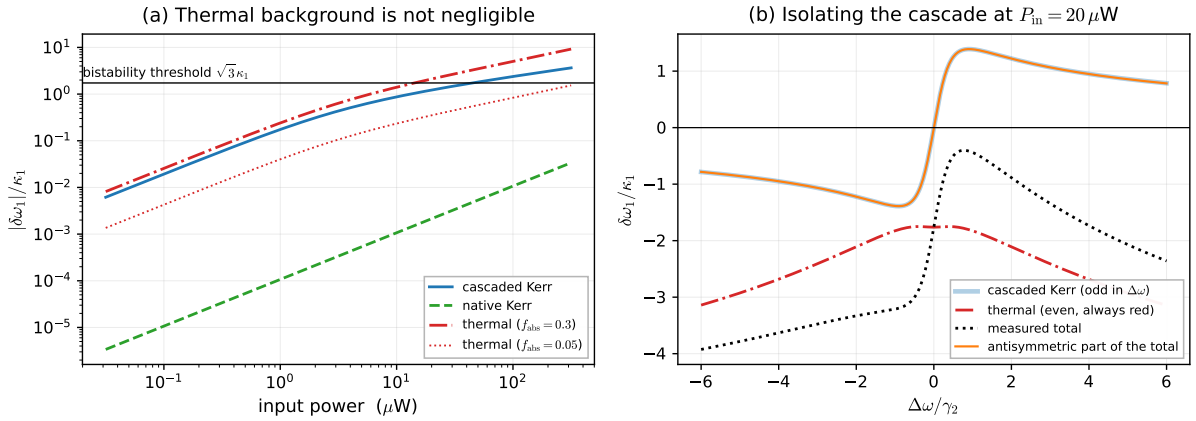


Figure 7. (a) Resonance shift versus input power for the cascaded Kerr, the native Kerr and the thermal background at two absorbed fractions. The cascaded curve bends over because at $\delta_1 = 0$ the shift detunes the cavity from its own pump. (b) At 20 μW the measured total (dotted) is thermally dominated, but its antisymmetric part (orange) reproduces the cascaded Kerr (thick blue) to 0.3%.

10 Recommended experiment

1. **Device.** A periodically poled LN microring, radius 30–70 μm, FM TM_{00} at 1.55 μm critically coupled ($Q_1^{\text{load}} \simeq 10^6$), SH TM_{00} at 775 nm left *under-coupled* (the cascade wants the SH stored, not extracted). Add electrodes for r_{33} tuning of $\Delta\omega$. A sister device with a deliberately out-coupled SH port is needed for step 2.

2. **Calibrate g.** Measure η_{SHG} in %/W on double resonance and invert (15). This fixes the absolute scale of every prediction in this note with no free parameters. Measure Q_1, Q_2 (loaded and intrinsic) in the same run.

3. **Map $\Delta\omega$.** Temperature-tune to bring $\Delta\omega$ through zero, identified by the maximum of SH power, and calibrate $d\Delta\omega/dT$ (~ 0.9 GHz/K) and $d\Delta\omega/dV$ against it.

4. **The sign-flip measurement.** At fixed pump power (10–50 μW), with the pump locked to the FM resonance ($\delta_1 = 0$, see Section 9), dither $\Delta\omega$ electro-optically between ± 2–$4\,\gamma_2$ at $\gtrsim 1$ MHz — far above the thermal and photorefractive response, and well inside the cascade's own 2.6 GHz bandwidth — and lock-in detect the transmission, re-referencing $\Delta\omega = 0$ to the SH power maximum. The in-phase component is $\propto \delta\omega_1^{\text{casc}}$ by (35). Sweep power to confirm linearity in n, and sweep $\Delta\omega$ to trace the full dispersive lineshape of Figure 3(a).

5. **Bistability.** Raise the power to $\sim 50\,\mu$W at $\Delta\omega \simeq +3.7\gamma_2$ and $\delta_1 < 0$; verify hysteresis, then reverse $\Delta\omega$ and verify the hysteresis appears on the other side of the resonance. This is the qualitative result no thermal mechanism can imitate.

6. **Control.** Detune the SH far away ($|\Delta\omega| \gg \gamma_2$) or temperature-tune off quasi-phase-matching, and repeat: the same device becomes the native-$\chi^{(3)}$ reference. *The control experiment is the same chip.*

The whole sequence needs microwatts, one tunable telecom laser, one temperature controller, and the electrodes that thin-film LN already has. No amplifier, no pulsed source, and — unlike the SH-only metasurface samples — no picosecond laser and no Z-scan.

11 Numerical validation

Everything above was checked against direct integration of (7)–(8), non-dimensionalised ($\tau = \kappa_1 t$, amplitudes in units of the linear-cavity amplitude) and including the native Kerr term $+iU_{\mathrm{nat}}|a|^2 a$. The transmission lineshape from the full two-mode integration agrees with the closed-form cubic (24) to 4×10^{-14} over the whole scan, which is machine precision for this problem. Additional checks, all reported by `ring_params.verify()`:

- $\eta_{\mathrm{SHG}} \leftrightarrow g$ inversion round trips to $< 10^{-12}$, and (15) matches the published Eq. (4) of Ref. [1] analytically.
- The linewidths recomputed from the quoted Q's of Ref. [1] reproduce their $\kappa_{a,0}/2\pi = 55.4$ MHz and $\kappa_{b,0}/2\pi = 343.8$ MHz to 3% (the residual is their quoted mode frequency of 199 THz versus $2\pi c/1560$ nm).
- The exact-versus-far-detuned ratio of U_{casc} matches $[1 + 1/(4x^2)]^{-1}$ identically for $x = \Delta\omega/\gamma_2$.
- The bistability condition (25) returns $|\delta_1| = \sqrt{3}\kappa_1$ at $\rho = 0$, and the fixed-coefficient cusp formula agrees with a direct minimisation of the fold power over δ_1 within that model to 0.1%.
- The 4×4 Bogoliubov matrix of Section 8.4 reproduces its analytic 2×2 adiabatic reduction (31) to 9×10^{-4} in the zero-walk-off limit, the residual being the finite-κ_2 correction to the elimination. (An earlier comparison appeared to fail by a factor of two; the discrepancy was the FSR walk-off, which the reduction ignores and the 4×4 keeps.)
- $n_2 \leftrightarrow \chi^{(3)}$ conversion round trips exactly.

What is *not* verified. The three-wave couplings are taken equal for every mode triplet; the thermal model is a lumped single-time-constant estimate with an assumed absorption partition; no photorefractive dynamics is modelled; and g for Design B is an extrapolation (Section 13).

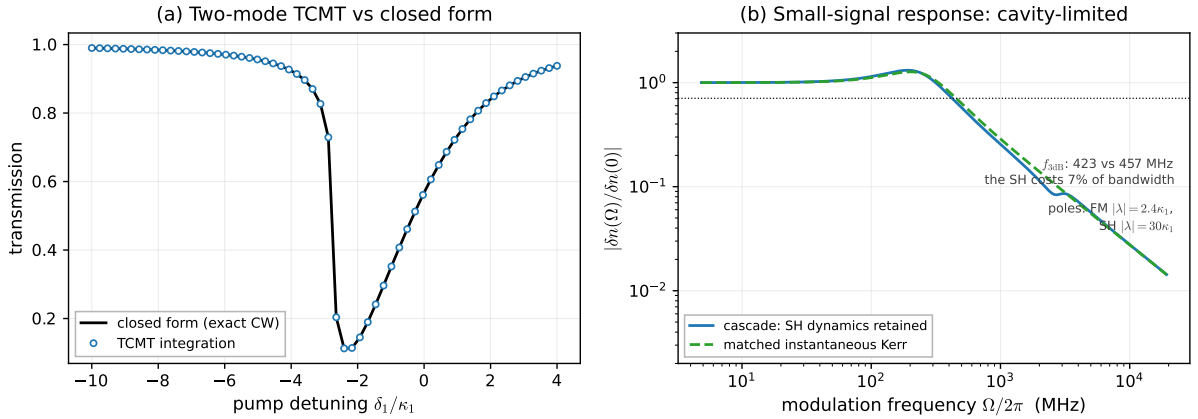


Figure 8. (a) Full two-mode TCMT integration against the closed-form steady state. (b) Frequency response of the cascaded Kerr compared with the FM cavity: the cascade rolls off at $\Delta\omega/2\pi = 2.6$ GHz, the matched instantaneous-Kerr reference: the SH's finite response time costs 7% of modulation bandwidth, so the resonator is the bottleneck (Rule 1).

12 Comparison across platforms and architectures

	metasurface (note §10)	SH-only samples	**PPLN microring**
Q_2	4×10^3	$0.9\text{–}5.6 \times 10^3$	$3\text{–}6 \times 10^5$
overlap mechanism	meta-atom	meta-atom	QPM over $2\pi R$
R at $\Delta\omega = 2\gamma_2$	~ 12	0.4–2.5	$\mathbf{1.8 \times 10^3}$
absolute scale known?	prefactor $\times 4$	$+\,C_{\text{bridge}}$	**yes, from** η_{SHG}
bandwidth bottleneck	SH ($\gamma_2 < \gamma_1$)	SH	**FM cavity**
CW Kerr device?	marginal	no (μrad)	**yes, at** μ**W**
sign flip observable?	yes	yes (ps Z-scan)	**yes, CW + lock-in**
principal risk	disorder	overlap	thermal, photorefr.

Table 3. The microring platform against the two metasurface platforms studied earlier in this project.

Against the *native*-$\chi^{(3)}$ architecture on the same ring, the cascade wins on every intensity-dependent observable by two to three orders of magnitude in threshold power, wins outright on sign tunability, very nearly breaks even on speed (the SH costs 7% of modulation bandwidth), and loses on three counts: it introduces a nonlinear loss channel that TPA-free LN does not otherwise have; it requires the doubly-resonant condition to be held to a fraction of γ_2; and it is mode-selective across a comb where the native Kerr is uniform.

13 Caveats and what would change the answer

In rough order of how much they could move the headline:

1. Q_2 **at 775 nm is assumed, not measured on this process** (3×10^5 loaded, 6×10^5 intrinsic, taken from Ref. [1]). Loaded values of this order near 780–800 nm are experimentally plausible and have precedent, but we do not claim they are routine, and the intrinsic ceiling in high-quality thin-film LN is higher than the value used here [3]. Since $R \propto Q_2$ this is the single largest lever: an order of magnitude down in Q_2 is an order of magnitude down in R, though even $Q_2 = 10^4$ leaves $R \simeq 61$. The floor is $Q_2^{\text{cross}} = 164$, below which the cascade loses to the native Kerr.

2. **Periodic poling must be available.** Without it R drops from $\sim 1.8 \times 10^3$ to ~ 13 (Table 2), thresholds rise into the milliwatt range where thermal effects dominate, and the experiment reverts to being as hard as the metasurface version. Everything in this note hinges on this one fabrication capability.

3. g **for Design B is extrapolated**, scaled as $V_{\text{eff}}^{-1/2}$ from the 70 μm device of Ref. [1] at fixed $A_{\text{eff}} = 1\,\mu\text{m}^2$. Poling fidelity and duty-cycle error degrade with radius, so this is optimistic at the $\sim 30\%$ level in g, i.e. up to $\sim 2\times$ in R. Step 2 of Section 10 measures it directly and removes the issue.

4. **The native LN Kerr is quoted at the pessimistic end.** We use $n_2 = 1.8 \times 10^{-19}$ m^2/W; reported values span $0.9\text{–}1.8 \times 10^{-19}$, and the smaller value would *raise* R by up to $2\times$. The choice is deliberately conservative.

5. **The thermal model is lumped and order-of-magnitude** ($R_{\text{th}} = 3 \times 10^4$ K/W, absorbed fraction 0.05–0.3 of the intrinsic loss, single time constant). It is used only to establish that the thermal background is comparable to or larger than the cascaded Kerr, which is robust across that whole parameter range; none of the cascade physics depends on it.

6. **Photorefractive drift is a real and unquantified risk** at exactly the $\gtrsim 10\,\mu$W powers of interest [1]. It acts on $\Delta\omega$ itself, so it corrupts the control variable rather than just the readout. Cladding removal [4], elevated temperature and MgO doping are the known mitigations; none is free.

7. **Comb applications need more than the two-mode model.** Section 8.4 establishes the parametric threshold rigorously, but the FM/SH free-spectral-range walk-off makes the cascade mode-selective, so a genuine comb calculation requires the multimode $\chi^{(2)}$ model, not the effective Kerr used here.

8. **The $\eta_{\mathrm{geom}} = 0.93$ implied by the metasurface note's formula sits at its physical ceiling** (Section 5), which points either to the larger β-prefactor branch or to an overestimated A_{eff}; as Section 4 says, a scalar estimate cannot separate the two. It does not affect any microring number in this note, all of which are anchored on the measured g, but it does need resolving before the two notes are quoted side by side.

9. **Equal three-wave couplings are assumed** across every mode triplet in the multimode calculation, and for the XPM factor of two. In a real ring $g_{12c} \neq g_{11b}$ because the frequencies, transverse profiles, tensor projections and QPM overlaps all differ.

14 Summary and recommendation

1. A doubly-resonant PPLN microring on the quoted platform ($Q_1^{\mathrm{load}} \sim 10^6$, $Q_1^{\mathrm{int}} \sim 2 \times 10^6$) with the demonstrated coupling $g/2\pi \simeq 1.2$–1.4 MHz yields $R \simeq 1.8 \times 10^3$ at $\Delta\omega = 2\gamma_2$, i.e. $n_2^{\mathrm{eff}} \simeq -3.3 \times 10^{-16}$ m^2/W. This is three orders of magnitude better than anything the metasurface route delivered, and the reason is Q_2 and quasi-phase matching, not luck.

2. **Periodic poling is the enabling step.** Without it, $R \simeq 13$ — the same order as the metasurface — and the crossover Q_2 rises from 164 to 2.4×10^4.

3. The absolute scale is measurable, not modelled: η_{SHG} in %/W fixes g through (15). This removes the C_{bridge} unknown that limited the SH-only sample analysis, and incidentally provides the first external evidence on the main note's β-prefactor ambiguity, favouring the larger branch.

4. Design rules: bistability and single-pair parametric gain both need the *pump-frame* ratio $\rho = \kappa_2/|\Delta_{\mathrm{eff}}|$ to satisfy $\rho < 1/\sqrt{3}$ — a condition on $\Delta_{\mathrm{eff}} = \Delta\omega - 2\delta_1$, which the experimenter controls, not a fabrication requirement on $\Delta\omega$; the Kerr description needs $\Omega_{\mathrm{env}} \ll \sqrt{\kappa_2^2 + \delta_2^2} = 27\kappa_1$ here; and the shift stays linear up to $|\delta\omega_1| \sim \Delta\omega$, which is $8Q_1/Q_2$ linewidths and therefore ample. The recommended cold-cavity operating point is $\Delta\omega \simeq 3\gamma_2$, not $2\gamma_2$.

5. Thresholds: bistability at 32–37 μW (native 14 mW), parametric oscillation at 8.4 μW from a multimode Bogoliubov calculation (native 9.4 mW), switching at 8.4 fJ intracavity, 13 μW of CW drive (native 15.4 pJ, 19 mW).

6. The cascade is a classical instrument, not a quantum one. Its dispersive anharmonicity is $\sim (g/\kappa_1)(g/\kappa_2)$; blockade needs resonant operation instead.

7. The binding constraints are thermal and photorefractive shifts, within a factor of two of the cascaded Kerr at these powers. To leading order both are even in $\Delta\omega$ where the cascade is odd, and LN's own Pockels effect tunes $\Delta\omega$ by $2.4\gamma_2$ per V/μm and far faster than either background responds. Photorefraction is not guaranteed to be exactly even, so any residual odd background must be bounded experimentally. The recommended first experiment is therefore an electro-optically dithered, antisymmetric-in-$\Delta\omega$ lock-in measurement of the FM resonance shift at 10–50 μW, with the same chip serving as its own native-$\chi^{(3)}$ control.

Recommendation. The microring platform is well suited to this physics — substantially better than the metasurface platform this project started from — and the decisive experiment is a CW, microwatt-level, room-temperature measurement on a single chip. The two open questions worth settling before committing are the SH quality factor actually achievable at 775 nm on this process, and whether periodic poling is available; the second is worth far more than the first.

A Reproducing the numbers

All numbers and figures come from two modules in `cascaded_chi3/`:

- `ring_params.py` — `RingParams` dataclass, the design points of Table 1, the closed forms (11)–(18), the $\eta_{\mathrm{SHG}} \leftrightarrow g$ inversion (15), the bistability algebra (25), and the thermal/electro-optic tuning of $\Delta\omega$. Running it prints the eight-part verification report of Section 11.
- `ring_effects.py` — the exact CW cubic, the non-dimensionalised two-mode TCMT integrator, the 4×4 linear-stability analysis, and the figure set in `cascaded_chi3/figures/ring/`.

Both run standalone: `conda run -n ml_base python ring_params.py`, then `ring_effects.py`. They are deliberately not wired into `run_all.py`, which builds the main-note figures only.

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